

SOEN 6011 D2: F6 Beta Function

From-Scratch Numerical Core, Tkinter GUI, Version Control, and Updated Requirements

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D2 Scope and Dependency Chain

D1 baseline

- Log-gamma identity selected
- Python TUI
- `math.lgamma` and `math.exp`

D2 increment

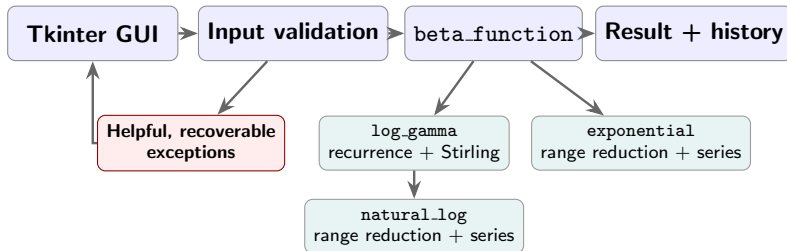
- 1 **P5:** implement mathematics from scratch
- 2 **P5:** replace TUI with Tkinter GUI
- 3 **P6:** publish Git repository + README
- 4 **P7:** revise requirements

Interpretation of “from scratch”

No `math` module or numerical library. Only input/output, arithmetic, exception handling, and Tkinter UI services are used. (Kamthan, 2026a)

D2 revisits the D1 solution and adds new behavior, following an iterative and incremental process. (Kamthan, 2026b)

P5: From-Scratch Architecture



Subordinate functions implemented from scratch: `natural_log`, `exponential`, and `log_gamma`.

Allowed dependencies

`tkinter`, `ttk`, float parsing, formatting, arithmetic.

Prohibited dependencies avoided

No `gamma`, `lgamma`, `log`, `exp`, `SciPy`, or `NumPy`.

P5: Numerical Method and Verification

Subordinate mathematics

$$\ln m = 2 \sum_{k=0}^{\infty} \frac{z^{2k+1}}{2k+1}, \quad z = \frac{m-1}{m+1}, \quad 1 \leq m < 2$$

$$e^r = \sum_{k=0}^{\infty} \frac{r^k}{k!}$$

Shift x to $x \geq 8$, then use the Gamma recurrence and a Stirling correction series.

$$\log B = \log \Gamma(x) + \log \Gamma(y) - \log \Gamma(x+y)$$

External verification only

Input	Computed value	Rel. error
$B(1, 1)$	0.999999999999988	1.24×10^{-14}
$B(2, 3)$	0.0833333333333322	1.45×10^{-14}
$B(0.5, 0.5)$	3.14159265358979	1.70×10^{-15}
$B(100, 300)$	$5.94745626861 \times 10^{-99}$	9.40×10^{-13}

Numeric limit

A representable finite result is returned; otherwise a `NumericRangeError` explains overflow or underflow.

All verification values were produced externally and used only for validation. The submitted implementation itself imports no mathematical library. (NIST DLMF, 2026a; NIST DLMF, 2026b)

P5: Tkinter GUI and Exception Recovery

Beta Function Calculator
Compute $B(x, y)$ for finite positive real inputs.

Inputs

x y

Result

$B(3, 2) = 0.0833333333333$
Calculation completed.

Calculation history

x	y	$B(x, y)$
3	2	0.0833333333333
2	3	0.0833333333333
1	1	1

Enter: calculate Esc: clear

GUI implemented entirely with Tkinter widgets. Demonstration layout shown.

User workflow

- 1 Enter labeled x and y
- 2 Calculate or press Enter
- 3 Read result and status
- 4 Swap, clear, or calculate again

Helpful exceptions

Missing: “ x is required.”

Nonnumeric: gives examples.

Domain: names field and > 0 rule.

Range: explains unrepresentable result.

Recovery

Errors update the status area; the window and history remain available.

P6: Distributed Version Control Evidence

High-quality local commit history

Commit	Purpose-focused message
30916e7	Implement from-scratch Beta numerical core
8d3270f	Add Tkinter GUI with recoverable input errors
1ba4daa	Document setup, numerical design, and demo cases
33ed8c8	Add repository hygiene and GUI demonstration evidence

README coverage

- Function and D2 scope
- Terminal run command
- From-scratch boundary
- Demo and error cases
- Numerical design and limits

Public repository

<https://github.com/HW925/SOEN-6011-D2-Beta-Function>

Commit logs and README files are lightweight software documentation that make decisions and usage explicit. (Kamthan, 2026c)

P7: Updated Functional Requirements

Legend: FR = Functional Requirement. These replace the D1 TUI-focused functional requirements.

GUI interaction and calculation

D2-FR-01 The system shall provide labeled GUI fields for x and y , plus Calculate, Swap, and Clear commands.

D2-FR-02 The system shall accept finite positive real inputs in decimal or scientific notation.

D2-FR-03 The system shall compute $B(x, y)$ through the from-scratch `natural_log`, `exponential`, and `log_gamma` operations.

D2-FR-04 The system shall display each representable result with at least ten significant digits.

D2-FR-05 The system shall preserve a history of successful calculations during the current session.

D2-FR-06 The system shall allow another calculation after a result or handled error without restarting.

Why requirements changed

D1's terminal prompts and quit command were replaced by GUI controls, persistent history, and recovery behavior required by P5.

P7: Updated Constraints and Quality Requirements

Constraints

D2-CR-01 The GUI shall use Tkinter.

D2-CR-02 The numerical core shall not call built-in or library mathematical functions.

D2-CR-03 The application shall run from a terminal with `python3 beta_function_gui.py`, independent of an IDE.

D2-CR-04 Each handled input error shall name the field and the correction needed.

Quality requirements

D2-QR-01 Relative error shall be at most 10^{-10} for $B(1, 1) = 1$ and $B(2, 3) = 1/12$; $B(2, 3)$ and $B(3, 2)$ shall agree within the same tolerance.

D2-QR-02 For $0 < x, y \leq 1000$ with a representable result, the calculation shall complete within 100 ms on the demonstration computer.

D2-QR-03 After a handled error, the GUI shall remain open and ready for corrected input.

Traceability: P5 architecture satisfies D2-CR-01–04; the verification table supports D2-QR-01; GUI recovery supports D2-FR-06 and D2-QR-03. (ISO/IEC/IEEE, 2018; Kamthan, 2026a)

GAI Use: CASTROFF Evidence for P5–P7

Tool and CASTROFF

Tool: OpenAI ChatGPT/Codex.

Context: SOEN 6011 D2, F6.

Audience: TA and instructor.

Style: concise, evidence-based.

Task: implement and document P5–P7.

Role: software engineering student.

Output: Python, README, requirements, slides.

Format: runnable files and LaTeX.

Feedback: verify mathematically and by execution.

Prompt examples and decisions

P5 prompt: “Design Beta from scratch without mathematical libraries, including subordinate functions and exceptions.”

Revision: chose recurrence + Stirling, added range checks, and rejected direct Gamma multiplication.

P6 prompt: “Propose purpose-focused commits and a minimal README for this scientific GUI.”

Accepted after check: four commits match actual repository changes; README commands were executed locally.

P7 prompt: “Revise D1 shall-statements for the D2 GUI and from-scratch constraint.”

Revision: separated functional, constraint, accuracy, timing, and recovery requirements.

References

Kamthan, P. (2026a). *SOEN 6011 Project Description*. Concordia University, Summer 2026.

Kamthan, P. (2026b). *Introduction to Software Processes*. SOEN 6011 lecture notes.

Kamthan, P. (2026c). *Introduction to Software Engineering Documentation*. SOEN 6011 lecture notes.

NIST Digital Library of Mathematical Functions. (2026a). Section 5.12, Beta Function, Release 1.2.7.

NIST Digital Library of Mathematical Functions. (2026b). Section 5.2, Gamma Function Definitions, Release 1.2.7.

ISO/IEC/IEEE. (2018). *29148:2018 Systems and Software Engineering – Life Cycle Processes – Requirements Engineering*.